

Deepak Sathyan

Articles in Review

3. **Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering** May 2026
B. Dutta, D. Goswami, J. Kumar, M. Rai, D. Sathyan
arxiv.org/abs/2605.08010
2. **A Baryon and Lepton Number Violation Model Testable at the LHC** Aug 2025
A. Bhoonah, F. Burk, D. Liu, T. Ou, D. Sathyan
arxiv.org/abs/2508.21064
1. **New Constraints on Neutrino-Dark Matter Interactions: A Comprehensive Analysis** July 2025
P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang
arxiv.org/abs/2507.01000

Articles in Refereed Journals

5. **New laboratory constraints on neutrinophilic mediators** July 2024
P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang
arxiv.org/abs/2407.12738 (Phys.Lett.B 868 (2025) 139765)
4. **‘Unification’ of BSM searches and SM measurements: the case of lepton+MET and m_W** Apr 2024
K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan
arxiv.org/abs/2404.17574 (JHEP 02 (2025) 139)
3. **A new purpose for the W-boson mass measurement: Searching for New Physics in lepton+MET** Oct 2023
K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan
arxiv.org/abs/2310.13687 (Phys.Lett.B 855 (2024) 138774)
2. **Energy-peak based method to measure top quark mass via B-hadron decay lengths** Dec 2022
K. Agashe, S. Airen, R. Franceschini, J. Incandela, D. Kim, D. Sathyan
arxiv.org/abs/2212.03929 (JHEP 06 (2023) 021)
1. **LHC Signals for KK Graviton from an Extended Warped Extra Dimension** Aug 2020
K. Agashe, M. Ekhterachian, D. Kim, D. Sathyan
arxiv.org/abs/2008.06480 (JHEP 11 (2020) 109)

Articles in Preparation

1. **LHC as a Beam Dump: Probing Light Dark Photons** Sept 2026
B. Dutta, A. Karthikeyan, D. Kim, H. Kim, T. Kim, D. Sathyan
arxiv.org/a/sathyan_d_1

- | | |
|---|----------|
| 2. New Probes of Dark Sectors at a Muon Collider
B. Dutta, A. Karthikeyan, D. Kim, K. Kelly, D. Sathyan
arxiv.org/a/sathyan_d_1 | Jan 2027 |
| 3. Exploring open systems approach to neutrino oscillations
A. Chandra Shekar, K. Kelly, D. Sathyan, L. Strigari, T. Zhou
arxiv.org/a/sathyan_d_1 | Mar 2027 |
| 4. LHC Signals for KK Gravitons Producing 8-Particle Final States
K. Agashe, D. Kim, P. Maksimovic, S. Mondal, M. Osherson, K. Panchal, D. Plotnikov, D. Sathyan
arxiv.org/a/sathyan_d_1 | Mar 2027 |

Ph.D. Thesis

- | | |
|---|------|
| 1. Unifying Searches for New Physics with Precision Measurements of the W Boson Mass
D. Sathyan
drum.lib.umd.edu/items/f9b081c8-444a-4af2-b69e-c8156d11a2c9 (Ph.D. Thesis) | 2024 |
|---|------|

Snowmass2021 Contributions

- | | |
|--|-----------|
| 5. TF07 Snowmass Report: Theory of Collider Phenomena
F. Maltoni et al.
arxiv.org/abs/2210.02591 | Oct 2022 |
| 4. Report of the Topical Group on Top quark physics and heavy flavor production for Snowmass 2021
K. Agashe et al.
arxiv.org/abs/2209.11267 | Sept 2022 |
| 3. Report of the Topical Group on Physics Beyond the Standard Model at Energy Frontier for Snowmass 2021
T. Bose et al.
arxiv.org/abs/2209.13128 | Sept 2022 |
| 2. Snowmass2021 - White Paper, Implications of Energy Peak for Collider Phenomenology: Top Quark Mass Determination and Beyond
K. Agashe, S. Airen, R. Franceschini, D. Kim, D. Sathyan
arxiv.org/abs/2204.02928 | Apr 2022 |
| 1. Snowmass2021 White Paper: Collider Physics Opportunities of Extended Warped Extra-Dimensional Models
K. Agashe, J.H. Collins, P. Du, M. Ekhterachian, S. Hong, D. Kim, R.K. Mishra, D. Sathyan
arxiv.org/abs/2203.13305 | Mar 2022 |

Muon g-2 Articles

3. **The straw tracking detector for the Fermilab Muon g-2 Experiment** Nov 2021
B.T. King et al.
arxiv.org/abs/2111.02076 (JINST 17 (2022) P02035)
2. **Beam dynamics corrections to the Run-1 measurement of the muon anomalous magnetic moment at Fermilab** Apr 2021
Muon g-2 Collaboration
arxiv.org/abs/2104.03240 (Phys.Rev.Accel.Beams 24 (2021) 044002)
1. **Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm** Apr 2021
Muon g-2 Collaboration
arxiv.org/abs/2104.03281 (Phys.Rev.Lett. 126 (2021) 141801)